

Laboratory report of Nanjing University of Information Science and Technology

Experiment Title virtualization Data 2026/5/21 Class 1
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一、Experimental purpose

- (1) Learn basic knowledge about AWS
- (2) Learn to use instances in AWS

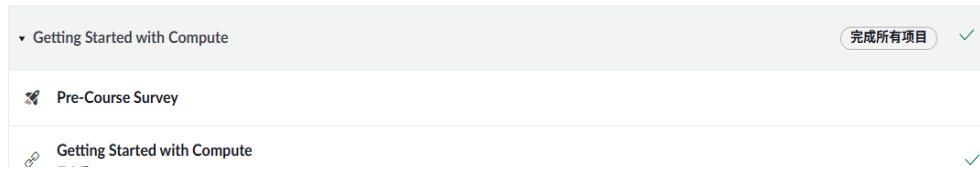
二、Experiment Contents

- Launch an EC2 instance with termination protection turned on.
- Monitor your EC2 instance.
- Modify the security group that your web server is using to allow HTTP access.
- Connect to your EC2 instance using AWS Systems Manager Fleet Manager.
- Manage the state of an EC2 instance.
- Change your EC2 instance type.
- Test termination protection.
- Explore Amazon EC2 limits.

三、Detailed steps of the experiment (please show the experiment steps and screenshots of the results)

Tasks:

1. [Watch: Getting Started with Compute](#)

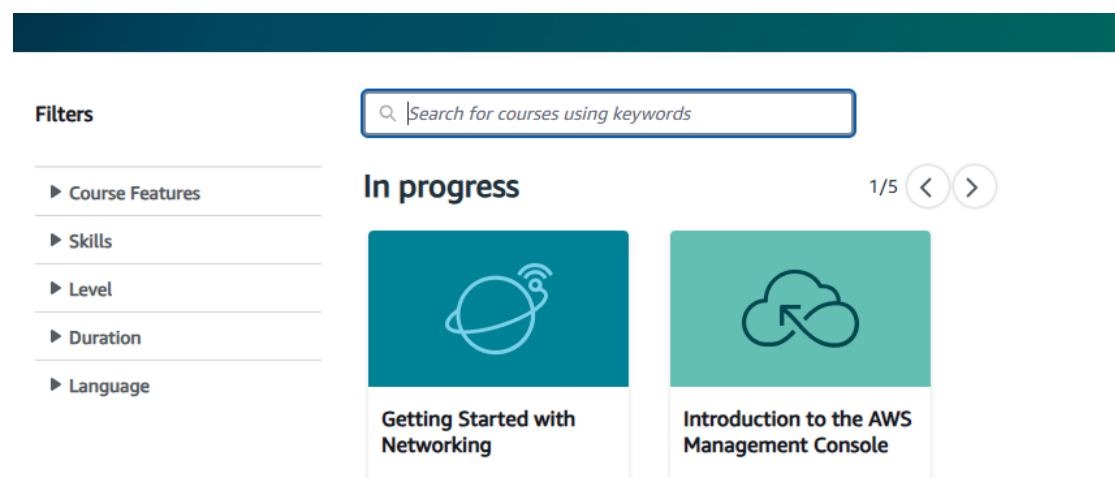


First, register with AWS Educate, then just click the link above.

If you cannot find the course from the above link, try the link below:

Learner Dashboard (<https://www.awseducate.com/student/s/content>) and

search the course: Getting Started with Compute



2. [Getting Started with Compute Simulation](#)



Please click the link above to complete your tasks. The experimental steps can be found in the instruction file attached.

四、Reflection

1. What is an EC2 instance essentially?

An EC2 instance is essentially a virtual server that runs in the AWS cloud.

It provides flexible and scalable computing resources for running

applications and services. Users can choose different operating systems,

resource sizes, and network configurations according to their needs.

2. What is the main advantage of server virtualization?

The main advantage of server virtualization is that it allows multiple virtual servers to run on a single physical machine. This greatly improves the utilization of hardware resources and reduces the cost of buying and maintaining physical servers. It also makes server management more flexible and efficient.

3. Why can cloud computing create servers quickly?

Cloud computing can create servers quickly because cloud platforms have a large pool of ready-to-use resources. The whole process is automated, so users do not need to install hardware or configure systems manually. With just a few settings, a new server can be launched and ready to use in minutes.

Task1:

COURSE MENU

Simulation Instructions ✓

Task 1

Launching your EC2 instance ✓

Task 2 ✓

Task 3

Task 4

Task 5

Updating your security group and accessing the web server

Connecting to your instance using AWS Systems Manager Fleet Manager

SIMULATION INSTRUCTIONS

Instances (1) info

Filter Instance by attribute or tag (enter keyword)

Name

Instance ID

Instance state

Instance type

Status check

Alarm status

Availability Zone

Public IPv4 DNS

Public IPv4...

Elastic IP

Web-Server

i-0a378ecc0300074da

Running

t2.micro

3/2 checks passed

View alarms

us-east-1a

ec2-54-89-233-131.co...

54.89.233.131

Select an instance

Task2

COURSE MENU

Simulation Instructions ✓

Task 1 ✓

Task 2

Monitor your instance ✓

Task 3 ✓

Task 4 ✓

Task 5

Task 6

Summary

Resizing your instance

Testing termination protection

SIMULATION INSTRUCTIONS

Instances (1/1) info

Filter Instance by attribute or tag (enter keyword)

Name

Instance ID

Instance state

Instance type

Status check

Alarm status

Availability Zone

Public IPv4 DNS

Public IPv4...

Elastic IP

Web-Server

i-0a378ecc0300074da

Running

t2.micro

3/2 checks passed

View alarms

us-east-1a

ec2-54-89-233-131.co...

54.89.233.131

Instance: i-0a378ecc0300074da (Web-Server)

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

Public IPv4 address

54.89.233.131

Open address

Private IPv4 address

10.0.1.12

Public IPv4 DNS

ec2-54-89-233-131.compute-1.amazonaws.com

Open address

Private IP DNS name (IPv4 only)

ip-10-0-1-12.ec2.internal

Subnet ID

subnet-0b16a33e0d8847ack

Public Subnet 1

Availability zone

us-east-1a

User data on guest OS

Download

Network Interface

eni-

Private IP address (ephemeral)

10.0.1.12

Carrier IP address (ephemeral)

Disabled

Amazon DNS Instance IPv4

Disabled

VPC ID

vpc-f0a6eab2156a74f18

Open VPC

Secondary private IPv4 addresses

-

Output ID

-

Filter network interface

Interface ID

Description

IPv4 Prefixes

IPv4 Prefixes

Public IPv4 address

Private IPv4 address

Private IPv4 DNS

IPv4 addresses

Primary IP

eni-

eni-0a378ecc0300074da

-

-

54.89.233.131

10.0.1.12

ip-10-0-1-12.ec2.inter...

-

-

Task3

COURSE MENU

Simulation Instructions

Task 1

Task 2

Task 3

Updating your security group and accessing the web server

Task 4

Task 5

Resizing your instance

Task 6

Testing termination protection

SIMULATION INSTRUCTIONS

Instances (1/1) info

Web-Server

i-0a378ec0300074da

Running

12.m3.xlarge

372 checks passed

View alarms

us-east-1a

ec2-54-89-233-131.eu...

54.89.233.131

Instance: i-0a378ec0300074da (Web-Server)

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

Instance summary

Instance ID

i-0a378ec0300074da (Web-Server)

Public IP address

54.89.233.131 (open address)

Instance state

Running

Private IP DNS name (IPv4 only)

ip-10-0-1-13.ec2.internal

Instance type

12.m3.xlarge

VPC ID

vpc-0a2a6eb255ba74f3 (Sub VPC)

Subnet ID

subnet-0b1a232db8476c3 (Public Subnet 1)

Private IP addresses

10.0.0.13

Public IP address

ec2-54-89-233-131.compute-1.amazonaws.com (open address)

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations. | Learn more

Auto Scaling Group name

Task4

COURSE MENU

Simulation Instructions

Task 1

Task 2

Task 3

Task 4

Connecting to your instance using AWS Systems Manager Fleet Manager

Task 5

Resizing your instance

Task 6

Testing termination protection

SIMULATION INSTRUCTIONS

EC2 Dashboard

EC2 Global View

Events

Console log

Permissions

Instances

Instance Types

Launch Templates

Spot Requests

Storage Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Tags

Images

AMI Catalog

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Network & Security

Security Groups

Elastic IP

Placement Groups

Key Pairs

Network Interfaces

Load Balancing

Load Balancers

Target Groups

Trust Stores

Auto Scaling

Auto Scaling Groups

Inbound security group rules successfully modified on security group sg-0732c055047233cb0 (Web Server security group)

Details

sg-0732c055047233cb0 - Web Server security group

Security group name

Web Server security group

Security group ID

sg-0732c055047233cb0

Description

Security group for the web server

VPC ID

vpc-0a2a6eb255ba74f3

Owner

929032431040

Inbound rules count

1 Permission entry

Outbound rules count

1 Permission entry

Inbound rules

Outbound rules

Tags

Inbound rules (1)

Name

sg-0a607f9ca1b4800b

Security group rule...

IPv4

Type

HTTP

Protocol

TCP

Port range

80

Source

0.0.0.0/0

Description

Task5

